

Refaat Elia

+20 1272360822 | refaatelia10@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Recent Computer Science graduate specializing in Artificial Intelligence with hands-on experience in machine learning, deep learning, computer vision, and Generative AI. Experienced in developing end-to-end AI systems using PyTorch, Vision Transformers, YOLO, FastAPI, Large Language Models, and Retrieval-Augmented Generation (RAG). Passionate about building practical AI solutions and continuously developing expertise in modern AI technologies.

EDUCATION

Ahram Canadian University

Bachelor of Science in Computer Science

Egypt

2022 – 2026

- **GPA: 3.43/4.00**
- Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Artificial Intelligence, Data Structures, Algorithms, Database Systems

EXPERIENCE & TRAINING

Large Language Models Intern

Sep. 2026 – Present

Tips Hindawi

Egypt

- Fine-tuned foundation models including Mistral, GPT, and Claude using Hugging Face Transformers for domain-specific tasks.
- Engineered Retrieval-Augmented Generation (RAG) pipelines leveraging LangChain, Vector Databases, and Python.
- Applied prompt engineering, context control, and model optimization techniques to improve output quality and latency.
- Built and deployed scalable, production-ready AI applications using Streamlit and Ngrok for live inference.

AI Agents & LLMs Trainee

Sep. 2026 – Present

Digital Hub (D-Hub) & Orange Digital Center

Egypt

- Building practical AI agents and LLM-powered applications using LangChain, Flowise, and modern LLM technologies.
- Learning prompt engineering, response control, memory and context management, and tool-using AI agents.
- Implementing Retrieval-Augmented Generation (RAG) with PDFs and Google Drive for knowledge integration.
- Developing multi-agent systems and AI automation workflows integrating emails, spreadsheets, APIs, and external tools.
- Building and deploying AI applications with web interfaces while learning debugging, optimization, and performance tuning.
- Developing AI agent projects that retrieve and summarize information and automate email and report generation.

Machine Learning Engineer Intern

2025

Digital Egypt Pioneers Initiative (DEPI)

Egypt

- Completed hands-on training covering Machine Learning, Deep Learning, and Computer Vision.
- Built machine learning pipelines including data preprocessing, feature engineering, model training, and evaluation.
- Worked with real-world datasets using Python, Scikit-learn, TensorFlow, and PyTorch.
- Applied machine learning workflows for model optimization, validation, and AI solution development.

PROJECTS

AI-Powered Chest X-Ray Diagnosis System | *PyTorch, Vision Transformers, FastAPI* Graduation Project

- Developed an end-to-end AI-powered medical imaging system for automated multi-label chest X-ray diagnosis.
- Trained a RAD-DINO Vision Transformer on more than 55,000 chest X-ray images to identify 11 thoracic abnormalities.
- Implemented Attention Rollout explainability to provide interpretable model predictions.
- Developed an Out-of-Distribution (OOD) Gatekeeper using MobileNetV3 to improve prediction reliability.
- Applied Test-Time Augmentation (TTA) to enhance inference robustness and prediction consistency.
- Designed an optimized preprocessing and inference pipeline for efficient medical image analysis.
- Developed a bilingual Retrieval-Augmented Generation (RAG) chatbot for answering medical imaging questions in Arabic and English.
- Integrated Large Language Models (LLMs) to generate bilingual radiology reports from model predictions.
- Deployed the AI system through FastAPI for real-time inference and web integration.

JobPilot — Multi-Agent AI Job Application Assistant | *Python, Hugging Face Transformers, FAISS, RAG, Gradio*

- Built a multi-agent AI system that analyzes CVs and job descriptions to identify skills, requirements, matching evidence, and candidate gaps.
- Implemented specialized AI agents for CV analysis, job analysis, candidate-job matching, application generation, interview preparation, and quality checking.
- Developed a RAG pipeline using FAISS and Hugging Face embeddings to retrieve relevant evidence from the candidate's CV and reduce unsupported claims.
- Used Hugging Face models locally for text generation and embeddings, with a Gradio interface for uploading documents and viewing results.
- Automated generation of personalized cover letters, application answers, interview questions, preparation plans, and a final PDF report.

Real-Time Object Detection for Autonomous Vehicles | *Python, YOLO, OpenCV*

- Developed a real-time object detection pipeline for autonomous driving applications.
- Optimized inference speed while maintaining accurate object detection performance.
- Evaluated detection robustness under varying lighting conditions and road environments.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C++, C#, SQL, HTML, CSS

Machine Learning: Scikit-learn, TensorFlow, PyTorch, Supervised Learning, Feature Engineering, Model Evaluation, Data Preprocessing

Deep Learning: CNNs, RNNs, Vision Transformers (ViT), Transfer Learning, Fine-Tuning, Neural Networks

Computer Vision: YOLO, OpenCV, Medical Image Analysis, Image Classification, Object Detection, Explainable AI, Attention Rollout

Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Hugging Face Transformers, Prompt Engineering

AI Agents: LangChain, Flowise, Tool-Using Agents, Agent Memory, Context Management, Multi-Agent Systems, API Integration

Data Science: NumPy, Pandas, Matplotlib, Seaborn

Frameworks: FastAPI, Flask, Streamlit, Gradio

Developer Tools: Git, GitHub, Docker, VS Code, Jupyter Notebook

LANGUAGES

Arabic: Native

English: C1